FATTY ACIDS: STRUCTURE, CLASSIFICATION AND FUNCTIONS

LECTURE OBJECTIVES

At the end of lecture the student should be able to:

- Define fatty acids
- Describe the structure of fatty acids
- Classify fatty acids on the basis of saturation, shape and nutrition
- Describe the functions of fatty acids
- Discuss the clinical significance of fatty acids in health and disease

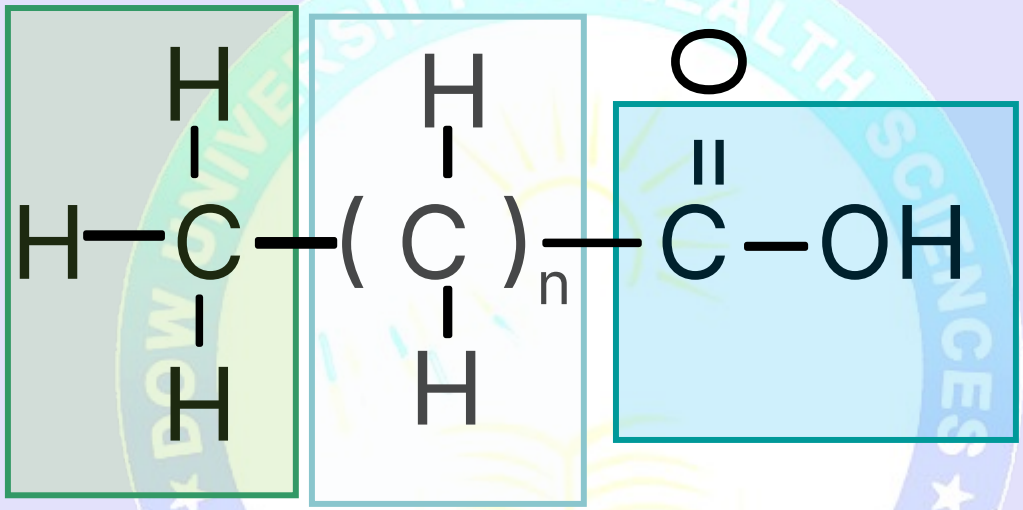
FATTY ACIDS

- Mono carboxylic acids with a hydrocarbon chain of varying length



- Hydrocarbon chain with **two** ends
  - Carboxylic acid
  - Methyl group

GENERAL STRUCTURE OF FATTY ACID



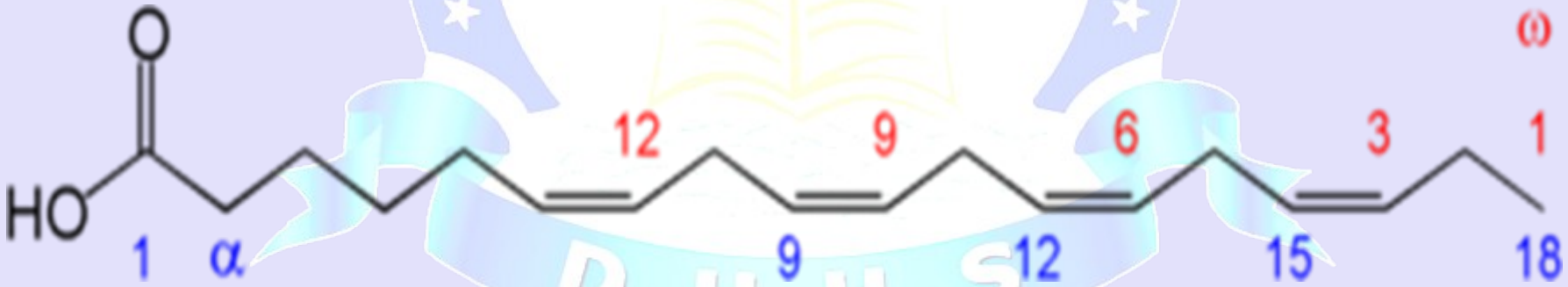
Methyl group

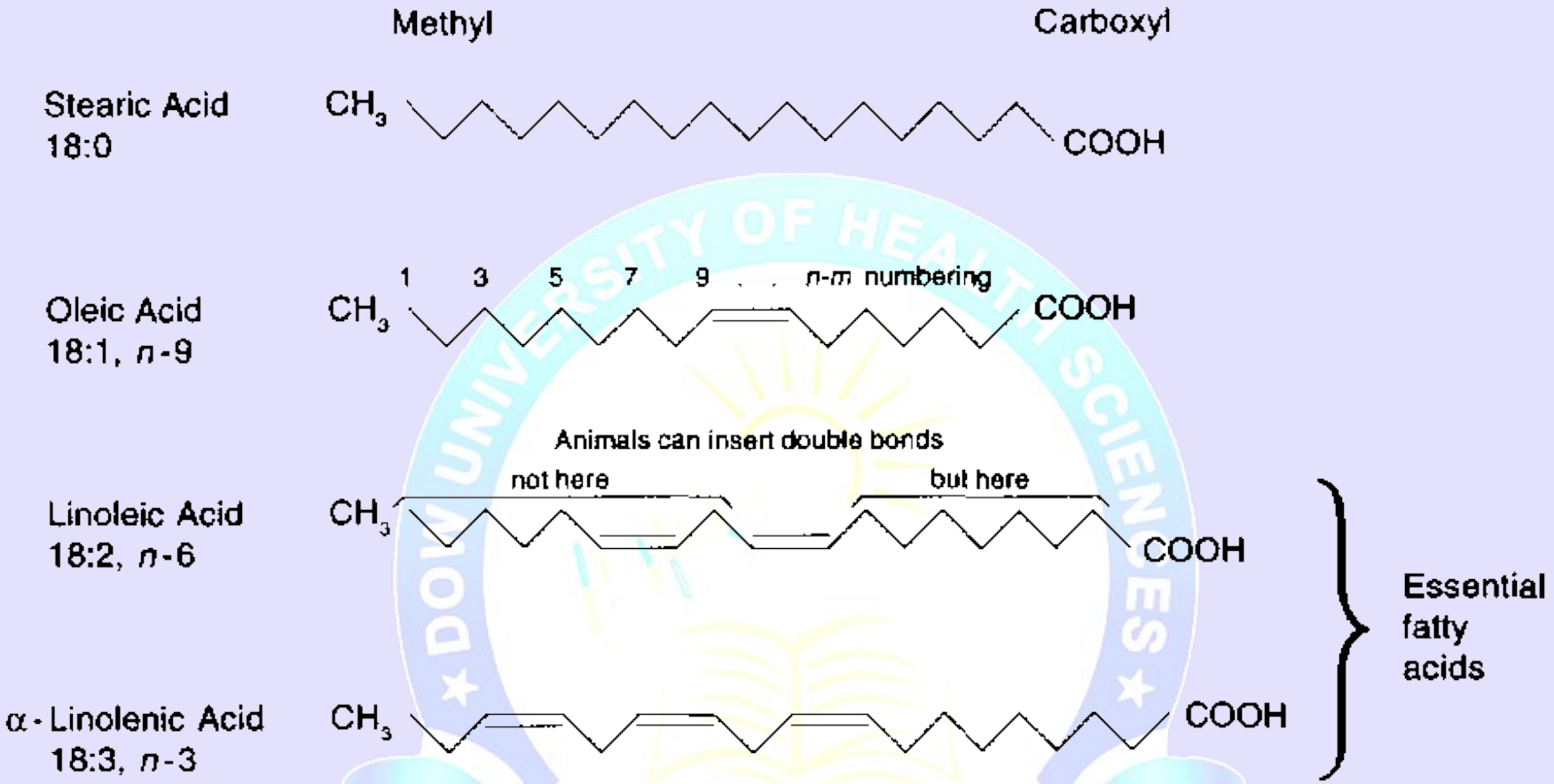
Carbon group(s)

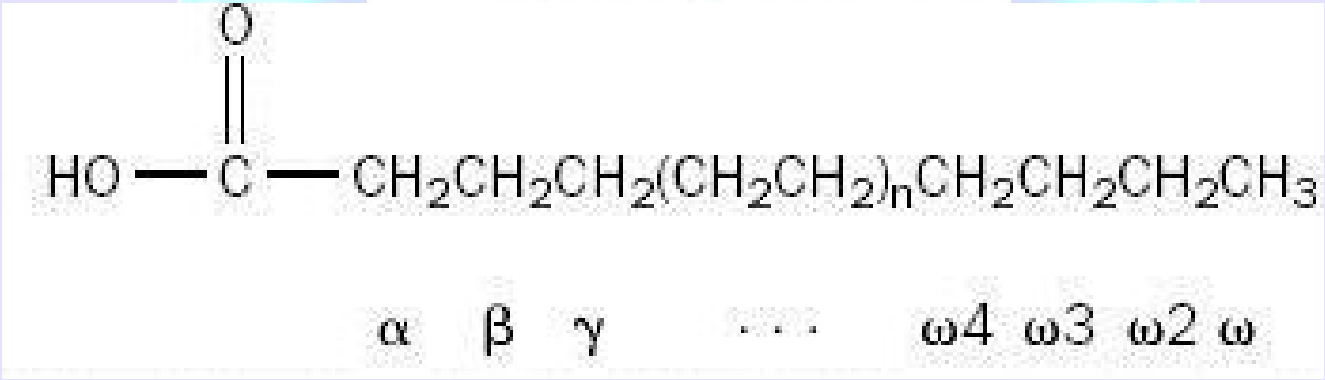
Carboxyl group

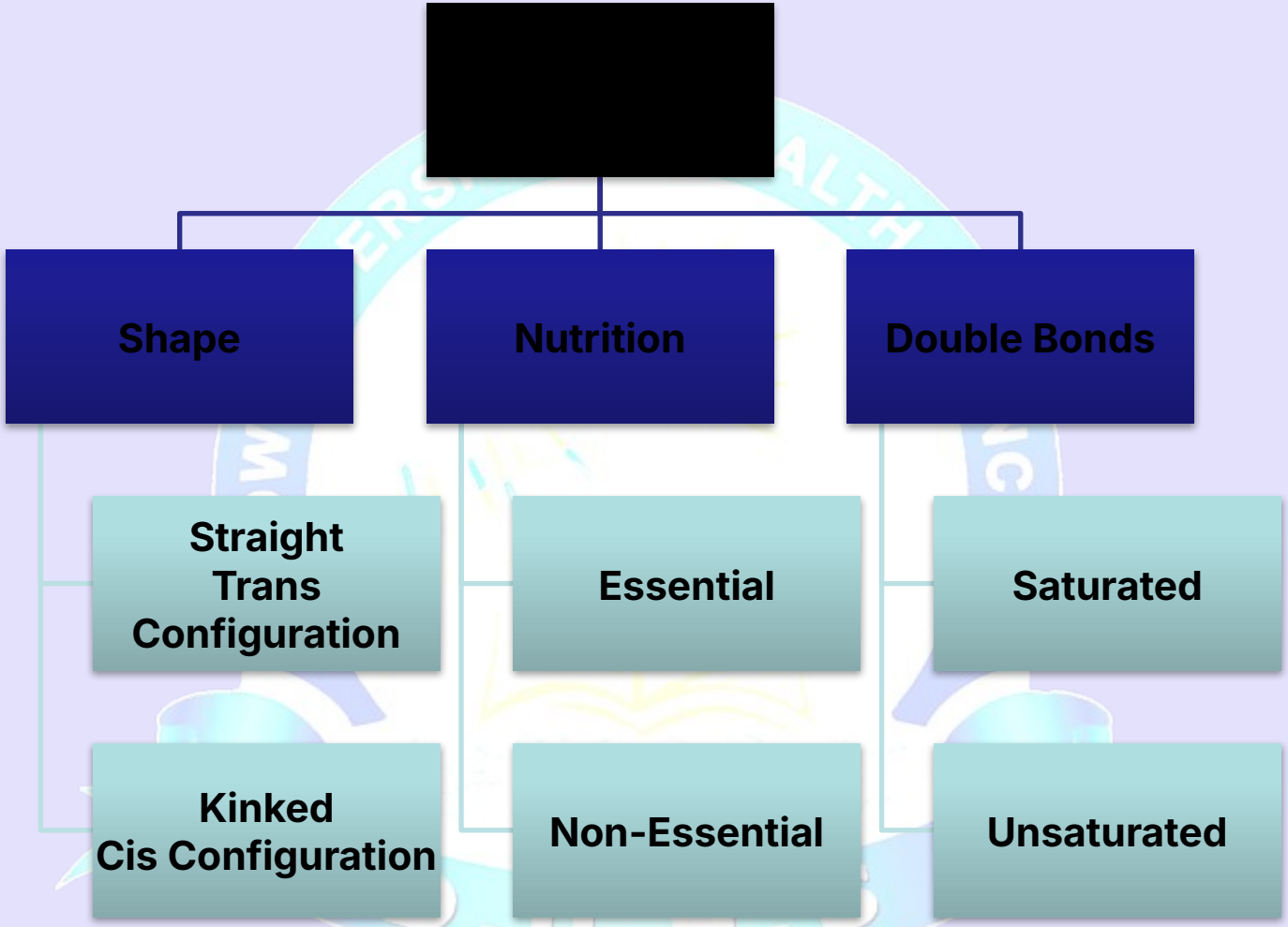
NOMENCLATURE

- Carbon next to carboxylic carbon is alpha (α)
- Terminal methyl carbon is termed omega (ω)




Methyl
Carboxyl
Stearic Acid
18:0
CH3
COOH
1 3 5 7 9 . . . n-m numbering
Oleic Acid
18:1, n-9
CH3
COOH
Animals can insert double bonds
not here
but here
Linoleic Acid
18:2, n-6
CH3
COOH
α-Linolenic Acid
18:3, n-3
CH3
COOH
Essential
fatty
acids


O
HO — C — CH2CH2CH2(CH2CH2)nCH2CH2CH2CH3
α β γ . . . ω4 ω3 ω2 ω


Shape
Nutrition
Double Bonds
Straight Trans Configuration
Kinked Cis Configuration
Essential
Non-Essential
Saturated
Unsaturated

FATTY ACIDS: SHAPES

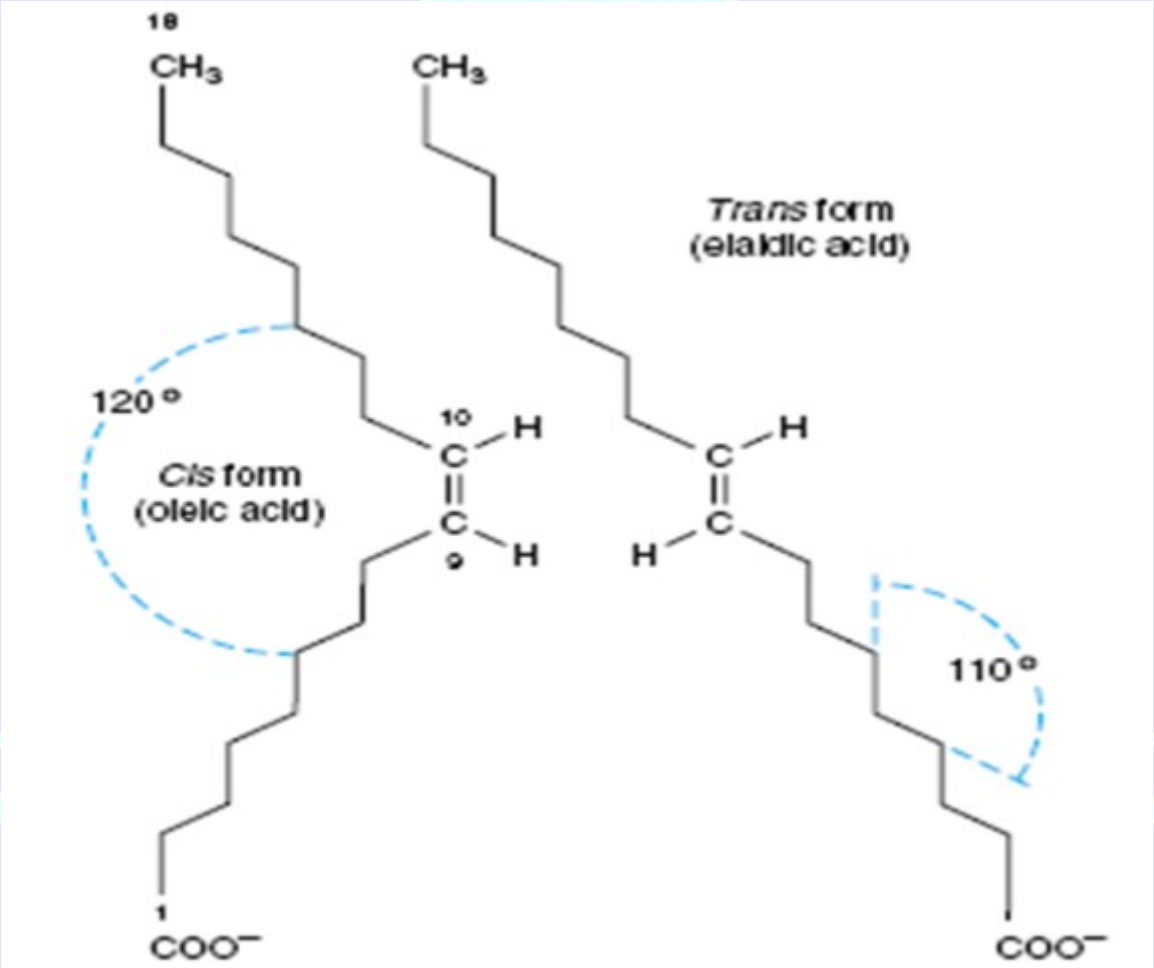
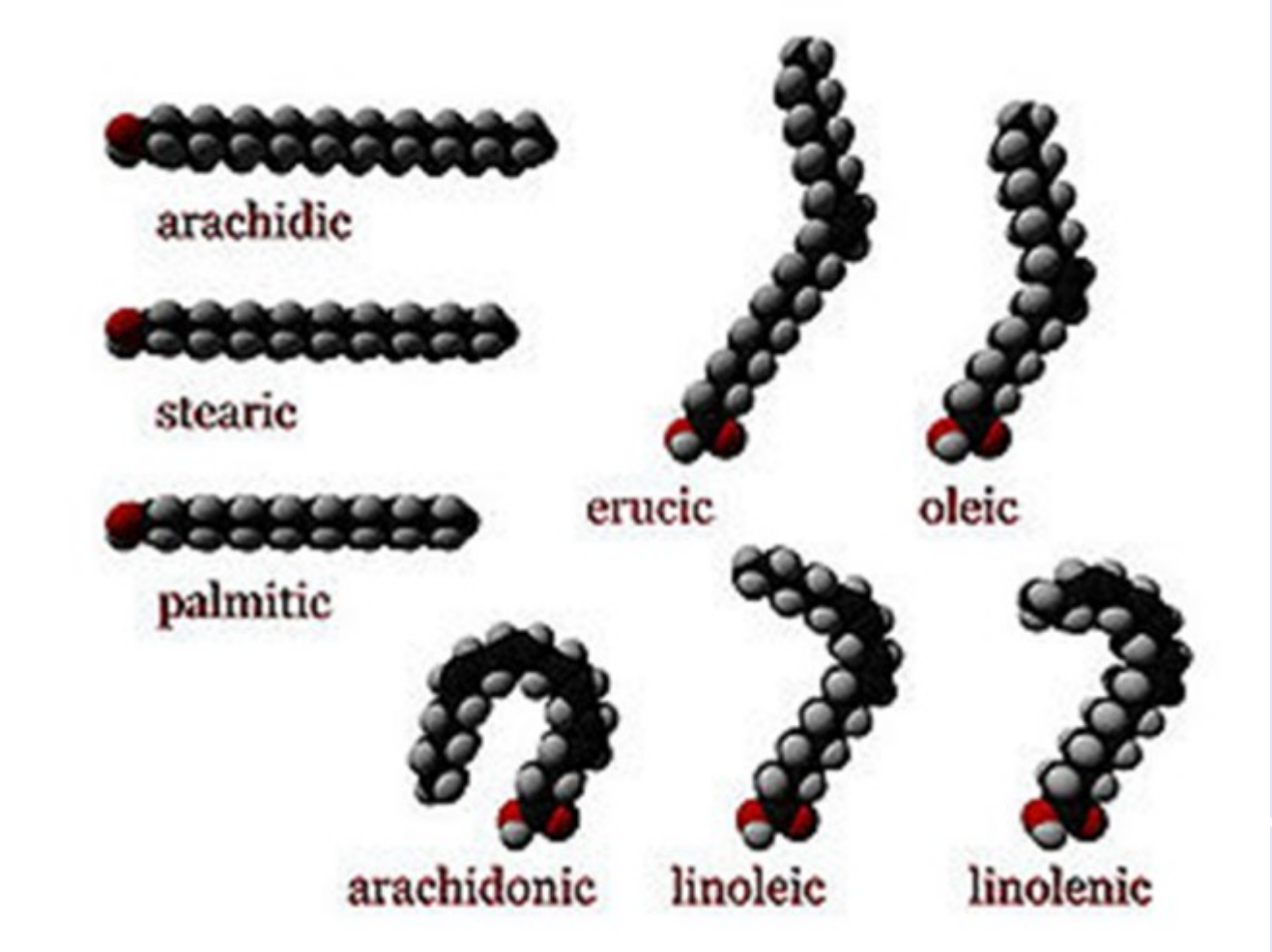


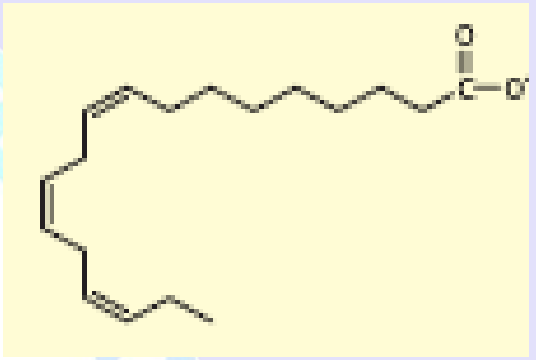
Figure 14–5. Geometric isomerism of Δ^9, 18:1 fatty acids (oleic and elaidic acids).

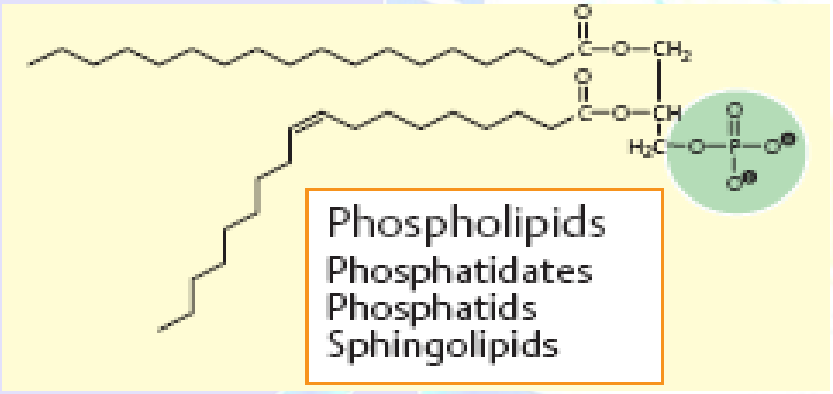
SOME FATTY ACIDS

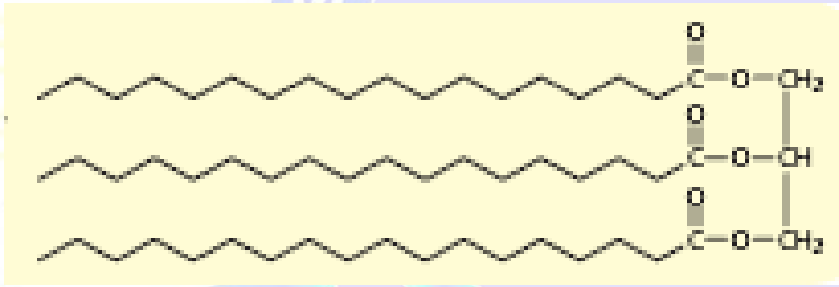


SOME FATTY ACIDS









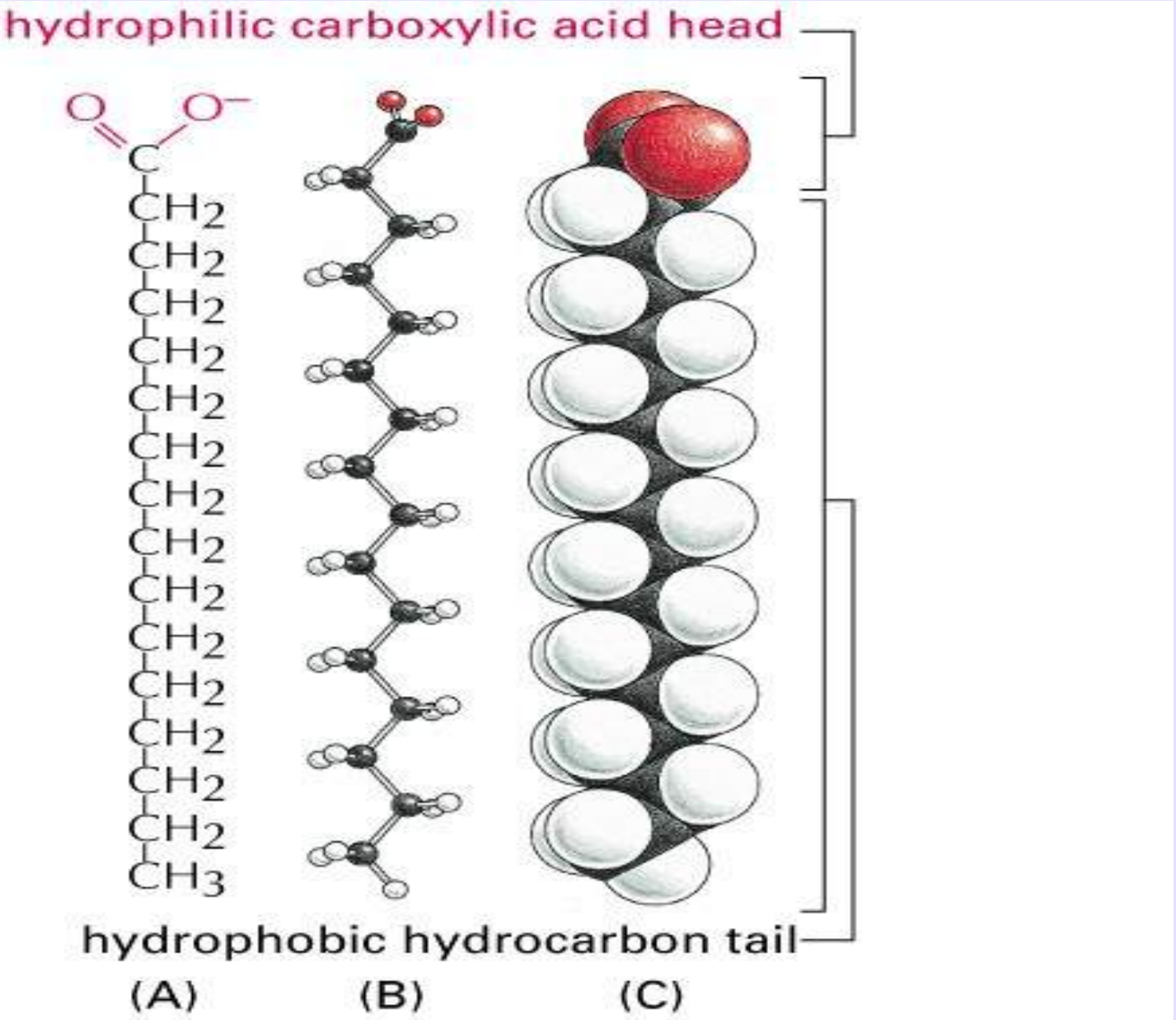


Figure 2–21. Molecular Biology of the Cell, 4th Edition.

UNSATURATED FATTY ACIDS

- Monounsaturated
- Polyunsaturated
- Ecosanoids
  - Derived from 20 carbon Fatty acid

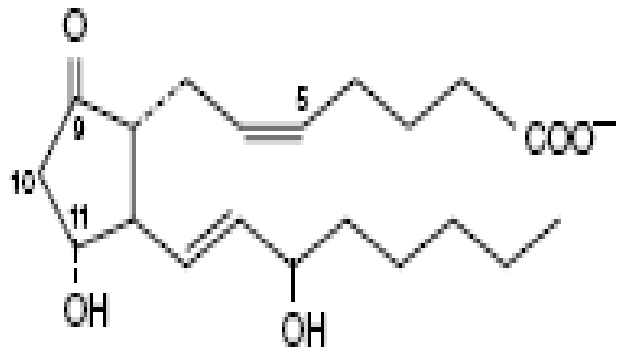


Figure 14–2. Prostaglandin E_2 (PGE_2).

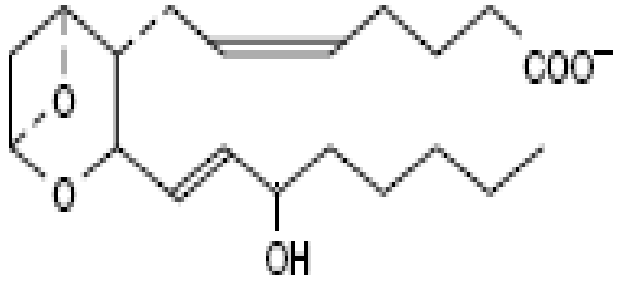
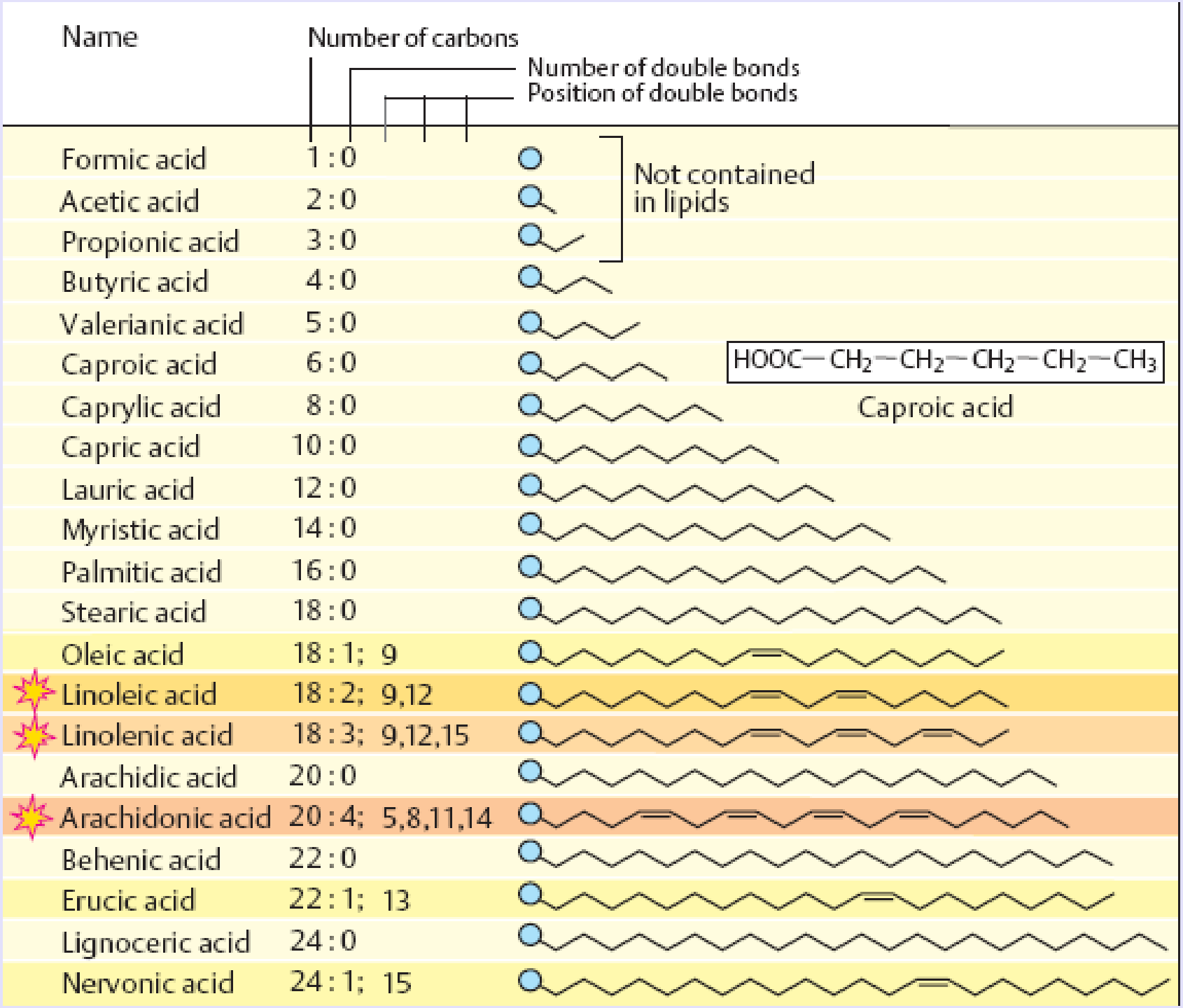
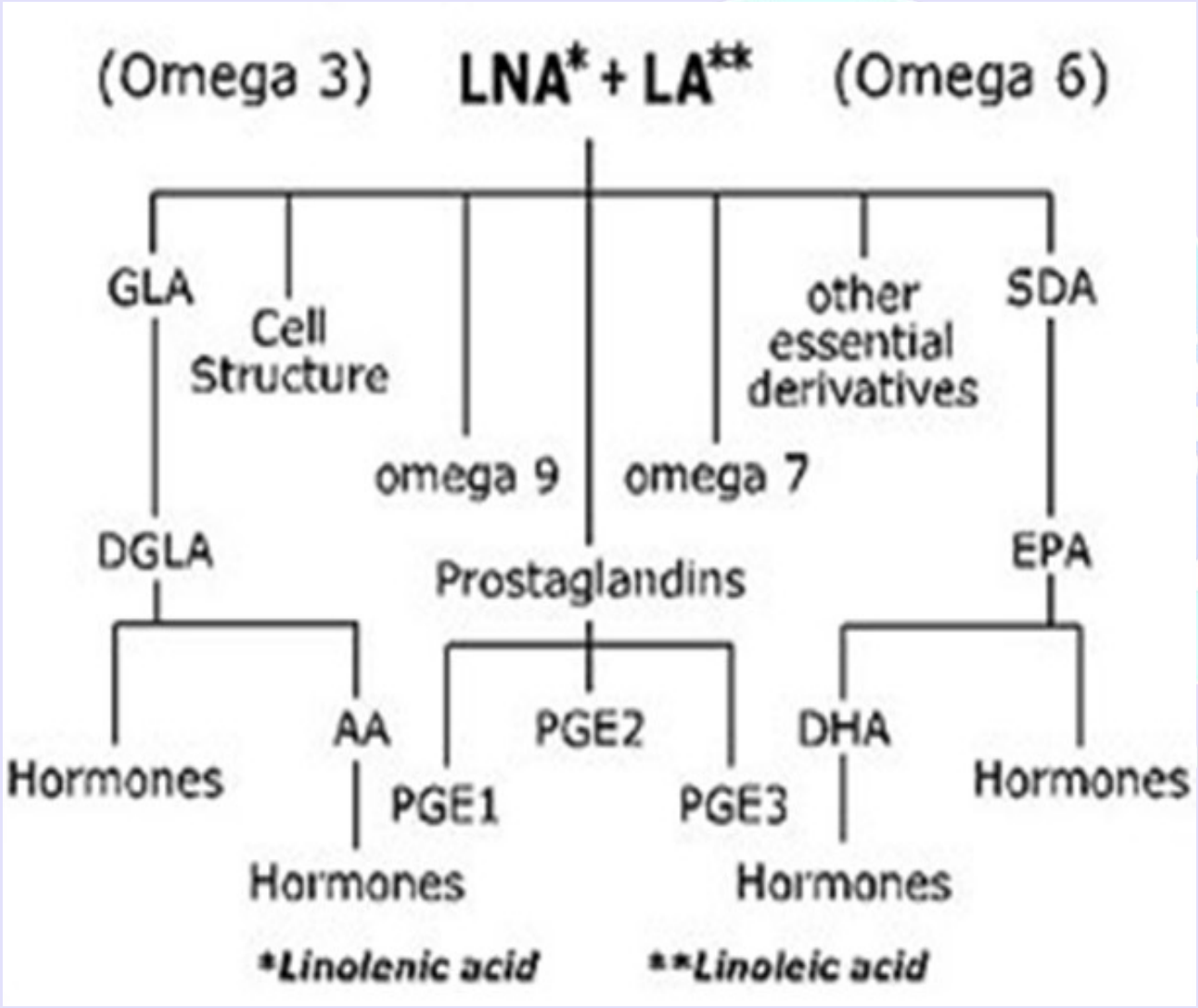


Figure 14–3. Thromboxane A_2 (TXA_2).

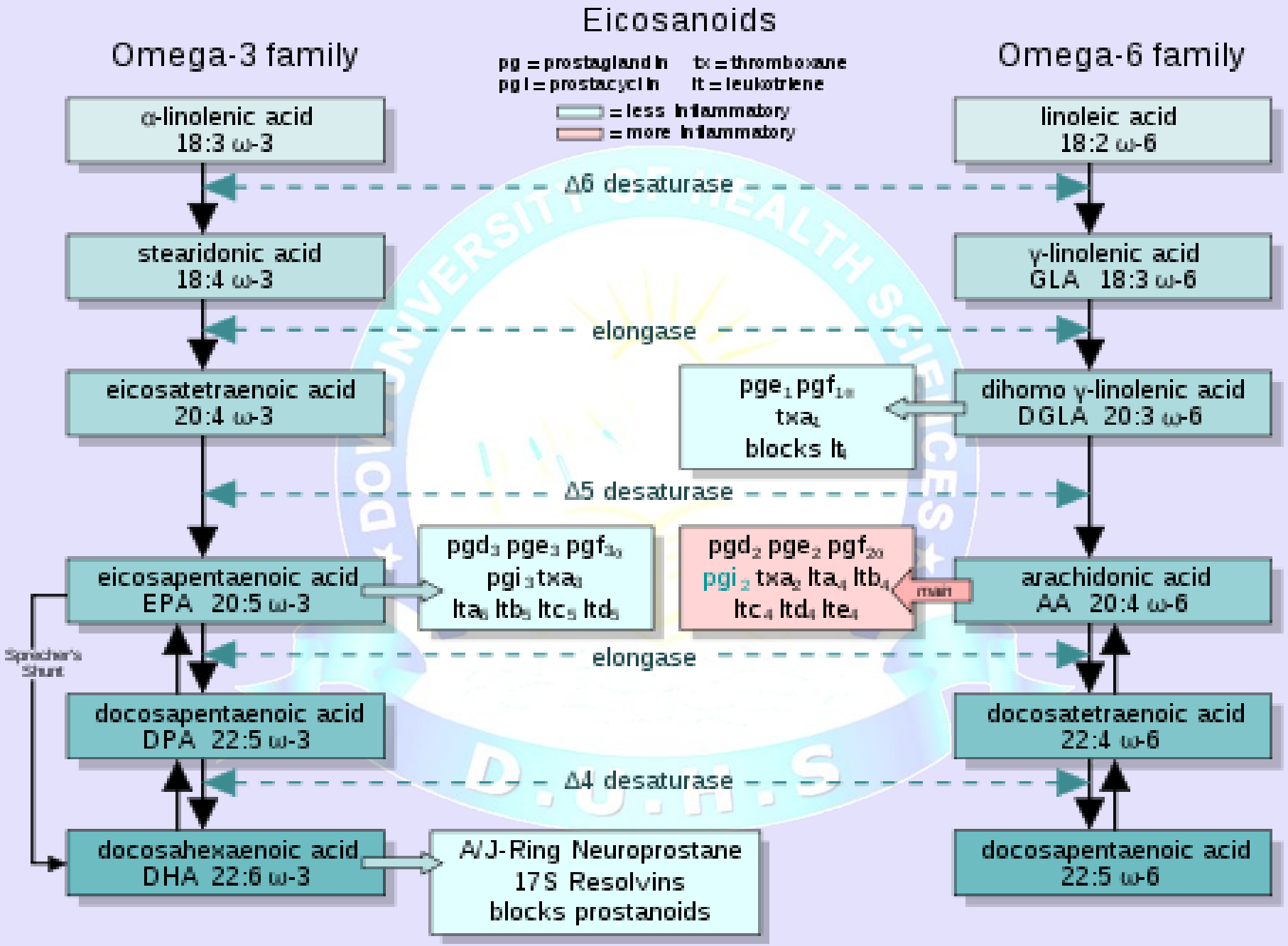

Name
Number of carbons
Number of double bonds
Position of double bonds
Formic acid 1 : 0
Acetic acid 2 : 0
Propionic acid 3 : 0
Butyric acid 4 : 0
Valerianic acid 5 : 0
Caproic acid 6 : 0
Caprylic acid 8 : 0
Capric acid 10 : 0
Lauric acid 12 : 0
Myristic acid 14 : 0
Palmitic acid 16 : 0
Stearic acid 18 : 0
Oleic acid 18 : 1; 9
Linoleic acid 18 : 2; 9,12
Linolenic acid 18 : 3; 9,12,15
Arachidic acid 20 : 0
Arachidonic acid 20 : 4; 5,8,11,14
Behenic acid 22 : 0
Erucic acid 22 : 1; 13
Lignoceric acid 24 : 0
Nervonic acid 24 : 1; 15
Not contained in lipids
HOOC—CH2—CH2—CH2—CH2—CH3
Caproic acid

ESSENTIAL FATTY ACIDS



SOME FUNCTIONS OF FATTY ACIDS

Molecule	Fatty acid	Function
Triacylglycerol	Saturated, unsaturated FA	Storage of energy Body contours
Vitamin D	Cholesterol	Calcium balance
Sex Hormones	Cholesterol	Regulatory functions
Triacylglycerol	Saturated fatty acids	Energy starvation
Local hormones	Eicosanoids., Prostaglandins	Inflammatory response, regulatory functions
Bile salts	Cholesterol	Lipid digestion and absorption
Phospholipids	Saturated, unsaturated FA	Membranes (cell, organelles, lipoproteins)
Sphingolipids	Saturated FA	Myelination


Eicosanoids
pg = prostaglandin
pgI = prostacyclin
tx = thromboxane
lt = leukotriene
= less inflammatory
= more inflammatory
Omega-3 family
Omega-6 family
α-linolenic acid
18:3 ω-3
linoleic acid
18:2 ω-6
Δ6 desaturase
stearidonic acid
18:4 ω-3
γ-linolenic acid
GLA 18:3 ω-6
elongase
eicosatetraenoic acid
20:4 ω-3
pge1 pgf1α
txa1
blocks lt4
dihomo γ-linolenic acid
DGLA 20:3 ω-6
Δ5 desaturase
eicosapentaenoic acid
EPA 20:5 ω-3
pgd3 pge3 pgf3α
pgi3 txa3
lta5 ltb5 ltc5 ltd5
pgd2 pge2 pgf2α
pgi2 txa2 lta4 ltb4
ltc4 ltd4 lte4
main
arachidonic acid
AA 20:4 ω-6
Sprecher's Shunt
elongase
docosapentaenoic acid
DPA 22:5 ω-3
docosatetraenoic acid
22:4 ω-6
Δ4 desaturase
docosahexaenoic acid
DHA 22:6 ω-3
A/J-Ring Neuroprostane
17S Resolvins
blocks prostanoids
docosapentaenoic acid
22:5 ω-6

CLINICAL SIGNIFICANCE

- Obesity
  - State of high adiposity
- Atherosclerosis
  - Cholesterol deposition narrows the arterial lumen and result into infarction of the target organ
  - Myocardial infarction
- High cholesterol content of the cell membranes limits the working of certain receptors like
  - Insulin receptors leading to Type 2 diabetes mellitus
- Decreased bile salts leads to deranged lipid digestion and absorption
  - Steatorrhea
- Omega 3 family of fatty acids are beneficial in hypertension

REFERENCES

- Lippincott illustrated review, biochemistry
  5th edition
  Ch 16, fatty acid & triacylglycerol metabolism
  Pg 181-182
- Harper's illustrated biochemistry
  28th edition
  Ch 15, lipid of physiologoic significance
  Pg 121-130